

VINAYAK SHARMA

BS Computer Science and Mathematics, UC San Diego; Incoming MS Computer Science student at Columbia University
Areas of Interest: Computer Science, Artificial Intelligence, Machine Learning, Math, Software Engineering, Cybersecurity
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EDUCATION

Bachelor of Science in Computer Science & Mathematics Sep'22 - Jun'26

- University of California, San Diego, CA (UCSD); GPA: 3.67/4.00, Provost Honors, Gary Reynolds Scholarship

International Baccalaureate Diploma (IBDP)

Aug'18 - May'22

- The International School Bangalore (TISB), India; GPA: 42/45

WORK EXPERIENCE

Scale AI, Inc., San Francisco, CA, GenAI Technical Advisor Intern (part-time) Sep'25 – Dec'25

- Improved LLM alignment and safety through evaluation of generative outputs, failure-mode analysis, and optimization of model accuracy and reliability.

Articul8 AI, Inc., Dublin, CA, Machine Learning Engineer Intern (full-time) Jun'25 – Sep'25

- Developed an AI pipeline for understanding complex tables, combining DETR-based table detection, Camelot parsing, and LLM-driven JSON schema generation to standardize hierarchical, merged, & nested table structures. Enhanced text-to-SQL accuracy to 95% using LLaMA few-shot learning & chain-of-thought prompting, enabling reliable data extraction across complex multi-table datasets. Built & integrated a custom GenAI agent into Articul8's AgentMesh platform, creating task-specific tools & workflows that achieved 100% functional parity with human performance for a Fortune 500 enterprise client.

MIT Lincoln Lab, Lexington, MA, AI Research Intern (full-time), Cyber-Physical Systems Jun'24 – Sep'24

- Built and evaluated a RAG system (>93% context precision) using RAGAS and Jaccard similarity. Experimented with document chunking strategies, prompt engineering, embedding models, retrieval methods, finetuning, and re-ranking.
- Created an LLM based 'SocialCipher' App with four other interns, to perform counter IO and generate threat scores for multimodal social media posts, achieving 96% accuracy. Presented a [paper](#) on this work at the IEEE CAI conference.

MIT Lincoln Lab, Lexington, MA, Summer Research Intern (full-time), Cyber-Physical Systems Jun'23 – Sep'23

- Investigated adversarial attacks against AI diffusion models through FGSM/PGD attacks to poison the input data, making them robust to attacks, and optimizing the diffusion models generation process by 20% through Progressive Distillation.

RESEARCH EXPERIENCE

Qualcomm Institute, UC San Diego, CA, AI / Computer Vision Undergraduate Researcher (under Neil Smith) Sep'25 - Present

- Built an LLM-powered NPC dialogue system using FastAPI, CouchDB, and Ollama for real-time generation and compliance-aware interactions in immersive environments. Integrated the pipeline into Unreal Engine via C++ components, enabling real-time API calls and scalable dialogue ingestion within interactive simulations.

SimWorld Research Project, UC San Diego, CA, Research Assistant (under Prof. Zhiting Hu & Prof. Lianhui Qin) Apr'26 - Present

- Running exploration experiments for improving embodied AI agents in realistic simulated physical and social environments.

Toor Lab, UC San Diego, CA, Machine Learning Researcher (under Prof. Navtej Toor) Jan'23 – Sep'24

- Trained and optimized NVIDIA GET3D model to generate 50+ 3D RNA high-fidelity models from 2D cryo-EM images.

Pioneer Research Institute, PA, Research Scholar (Mentor: Prof. S. Uludag, Univ. of Michigan) Jun'21 - Sep'21

- Conducted research on detecting zero-day cyberattacks using deep learning (LSTM) and achieved 97% accuracy ([Link](#)).

PROJECT EXPERIENCE

Triton Software Engineering (TSE), UC San Diego, CA, Product Manager, Software Developer Nov'22 – Jun'25

- Led a SW development team (13 people) for the non-profit 'HoMEwork' serving 100+ formerly incarcerated individuals; drove SW requirements/priorities using Agile for 30+ features and achieved 100% spec. completion ahead of schedule.
- Front-end developer for web app and Android mobile app React projects 'Laktaa Bhutan' and 'DFM Sideline Sidekick'. Developed 15+ production-ready UI components. Reviewed 35+ PRs leading to ~25% fewer QA issues in test builds.

CSForEach TritonHacks, UC San Diego, CA, External Events Coordinator Nov'23 – May'24

- Led a 6-person team to organize a hackathon for high school students, and secured company sponsorships for funding.

PUBLICATIONS

T. Pope, C. Lemieux-Mack, V. Sharma, A. Ogedengbe, A. Thanvi, "SocialCipher: A Multimodal Framework for Proactive Threat Detection with SentIntel" ([Link](#)), *IEEE Conference on Artificial Intelligence (CAI)*, 2025. May'25

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, TypeScript; **ML & Data Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, NLTK, OpenCV, Hugging Face; **AI/ML Frameworks & Tools:** LangChain, RAG, FAISS, Weights & Biases, NVIDIA GET3D, Adversarial training, Knowledge retrieval, Prompt engineering; **Cloud & SW Dev. Tools:** GCP, AWS, Colab, Jupyter Notebook, Docker, Git, GitHub, Visual Studio Code.